

Tensor computations

Infinite mean condition number in every identifiable tensor format

Difficulty: challenging **Importance:** interesting to the community**Status:** Partially resolved

Rating rationale: Challenging because extending the mean-divergence theorem requires controlling singular decomposition geometry beyond the existing identifiability hypothesis; community importance comes from average numerical conditioning of CP decomposition.

Context and notation

Fix $d \geq 3$, $n_j \geq 2$, and $r \geq 2$. Assume generic complex identifiability: outside a proper algebraic exceptional set, a complex tensor of rank r in this format has a unique unordered collection of r rank-one summands. Let M_r be the smooth identifiable locus of real rank- r tensors, with induced Euclidean volume dV . The random input has density

$$d\mu(A) = Z^{-1} e^{-\|A\|_F^2/2} dV(A).$$

For the addition map $\Phi(a_1, \dots, a_r) = \sum_i a_i$ on rank-one tensors, let Ψ be a local inverse at A . Use Frobenius norms and their product norm. Define

$$\kappa(A) = \|D\Psi(A)\|_2, \quad \kappa_{\text{ang}}(A) = \|D(p^{\times r} \circ \Psi)(A)\|_2, \quad p(a) = a/\|a\|_F.$$

Values on measure-zero exceptional sets do not affect the expectations. This samples tensors by volume, not independent Gaussian summands.

Problem statement

Under this probability model, prove or disprove

$$\mathbb{E}_\mu \kappa(A) = \infty$$

for every admissible format and $r \geq 2$. Rank two is already proved. The unresolved task removes the additional smaller-format identifiability assumption used for the higher-rank theorem; it must not be counted again for parameter ranges covered by that theorem.

Reference

Carlos Beltrán, Paul Breiding, and Nick Vannieuwenhoven, *The Average Condition Number of Most Tensor Rank Decomposition Problems Is Infinite*, *Foundations of Computational Mathematics* 23 (2023), 433–491: Definition 1, Assumption 1, equation (5), Theorems 1–2, and Conjecture 1. The [author preprint](#) labels the conjecture 1.8.

Status check — 2026-09-10

Rechecked the [journal version](#), [Theorems 1–2](#) and [Conjecture 1](#), and searched for subsequent average-condition-number resolutions. Rank two and the higher-rank formats satisfying the theorem’s extra smaller-format identifiability assumption are proved portions of the displayed family; removing that assumption remains unresolved. No later general proof was located. The latest explicit conjecture checked remains the 2023 journal version, so the later-status evidence is a bounded search.